NOT MEASUREMENT SENSITIVE

MIL-S- 83829
26 OCTOBER 1992

SWEEPER, ROTARY, SNOW, FRONT MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers a rotary, front mounted type snow sweeper that is diesel-engine-powered and mounts on a dump truck; the rotary broom on the vehicle's front, the diesel engine on skids in the dump body and the blower at the rear of the vehicle.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS
MILITARY

MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
MIL-H-83282	- Hydraulic Fluid, Fire Resistant, Synthetic, Hydrocarbon Base, Aircraft.
MIL-F-83828	- Fillers and Spacers for Rotary Sweeper Brushes, Front Mounted or Towed Type Sweepers.

STANDARDS
FEDERAL

FED-STD-595	- Colors.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Warner Robins Air Logistics Center, ATTN: WR-ALC/LVRG, 225 Ocmulgee Ct, Robins AFB GA 31098-1647, by using the self-addressed Standarization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-831	- Test Reports, Preparation of.
MIL-STD-965	- Parts Control Program.
MIL-STD-1474	- Noise Limits for Army Materiel.
MIL-STD-1556	- Government/Industry Data Exchange Program Contractor Participation Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DEPARTMENT OF DEFENSE (DOD)

Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093).

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulations.

(Application for copies of DOT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington D.C. 20402.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D1655	- Standard Specification for Aviation Turbine Fuels.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

B2.1-84	- Welding Procedure and Performance Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

J88	- Sound Measurement, Earthmoving Machinery, Exterior.

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J534	- Lubrication Fittings.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation From Vehicles and Devices (30-1000 MHz).
J578	- Color Specification.
J845	- 360 Degree Warning Lamp for Authorized Emergency, Maintenance and Service Vehicles.
J1349	- Engine Power Test Code, Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in the specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish a complete truck mounted snow sweeper for first article testing in accordance with sections 4 and 6.

3.2 Commerciality. The snow sweeper, accessories, and attachments shall be the manufacturer's standard commercial (see 6.5.1) product, except as modified to meet the requirements of this specification. The sweeper shall have been manufactured and sold to the general public at least two years prior to the opening date of the solicitation. Upon request of the contracting officer, offerers shall submit evidence to establish commerciality of their machines in the form of catalogs, commercial brochures and data. Additionally, these bidders shall furnish names and addresses of non-Government sources to which they have sold this type sweeper at least two years prior to the opening date of the solicitation.

3.2.1 Parts control. The contractor shall participate in the Government Industry Data Exchange Program (GIDEP) in accordance with MIL-STD-1556. In the design of the vehicle, the contractor shall ensure the components selected have not been identified as a problem or unresolved deficiency by GIDEP. The contractor shall submit an Alert/Safe-Alert on a DD Form 1939 whenever an actual or potential problem is evident with respect to a part, component, material, process, safety condition or to test equipment. The contractor shall certify maximum use of off-the-shelf items in accordance with MIL-STD-965.

3.3 Safety. All rotating or reciprocating parts, and all parts subject to high operational temperatures that are of such nature or so located as to be hazardous to personnel shall be guarded or insulated to eliminate the hazard. Guards and insulation shall have no effect on the operational functions of the sweeper. A warning/caution marking shall identify potential hazards such as pinch points, air-blast, etc. The marking shall include the word "warning" or the word "caution" (i.e. "warning pinch point"). The marking shall be applied using reflectorized tape and shall consist of one inch red letters on a silver background. A ladder shall be provided on the equipment to assist personnel in entering the dump bed of the truck to service the engine. The hydraulic component burst pressure rating must be a minimum of a 150 percent of the rated working pressure for the system. Hydraulic system components must meet or

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exceed the working pressure ratings. The engine skid and blower assemblies shall be secured by positive means to prevent them from shifting in the dump body.

3.4 Maintainability. Components of the sweeper shall be readily accessible for repair or replacement, with minimum removal or disturbance of adjacent parts or components using general purpose tools. The auxiliary engine, fuel tanks, hydraulic components and blower assembly shall be capable of being serviced or repaired while mounted in the back of the dump truck. Operating parts, accessories and drain outlets shall be easily accessible for regular maintenance service. Engine oil, fuel and coolant outlets shal permit draining of the fluid into a pan or pail without dripping on any part of the vehicle or its mounted equipment. Fluid reservoir filler points shall be readily accessible for checking fluid levels and adding fluid without having to remove components. Use of pressurized fluid dispensing systems shall not be acceptable to satisfy this requirement. All wear type parts shall be capable of being replaced, e.g. bushings, etc.

3.4.1 Truck Servicing. When the complete sweeper assembly is mounted on the truck, no component of the sweeper assembly shall interfere with servicing or maintenance of the truck. For example, the truck engine hood shall be capable of fully opening with the sweeper installed, even if it is the "tilt-forward" type.

3.4.2 Maintenance tasks. The design of the unit shall allow the following maintenance tasks to be performed by one person using common hand tools, found in an Air Force vehicle technician's individual tool kit, within the specified time limits. No electrical or pneumatic power tools or equipment shall be used to perform any of these actions. The engine skid assembly must be mounted in the back of a dump truck when these tasks are performed.

a. Drain engine oil, remove and replace engine oil filter(s), refill crankcase and check for oil leaks and proper oil level; grease the unit following the manufacturer's lubrication instructions; clean, service or replace engine air filter; - 1 hour.

b. Replace engine fuel filter(s); bleed system as necessary to remove any entrapped air; start engine and check for leaks - 1/2 hour.

c. Drain radiator, replace upper and lower radiator hoses, refill radiator to the proper operating level and check for leaks - 3/4 hour.

d. Drain hydraulic system; replace all hydraulic fluid filters and refill system - 1 hour.

3.4.3 Hydraulic system sensors. Hydraulic sensors compatible with UCC monitor, part number US-S20M-6526411111, shall be installed in the brush and blower drive circuits so that pressure, flow rate and temperature can be monitored without disconnecting any piping or tubing. All hydraulic fluid shall be in accordance with MIL-H-83282.

3.5 Painting. The sweeper shall be treated, primed and painted in accordance with the manufacturer's standard commercial (see 6.5.1) practice. The finish coat shall be the manufacturer's standard commercial (see 6.5.1) polyurethane or acrylic urethane paint. The finish color shall be Forest Green, Color Number 24052 of FED-STD-595.

3.6 Identification marking. A corrosion resistant metal nameplate,

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permanently and legibly marked with the following information in accordance with MIL-STD-130 shall be securely attached to each sweeper:

a. Nomenclature.

b. Manufacturer's model number.

c. Manufacturer's serial number.

d. Manufacturer's name.

e. Contract number.

f. Date of manufacture.

g. Special characteristics, HP, RPM, etc.

h. Weight.

i. Cube and center of gravity.

3.7 Lubrication. All moving parts and pivot points which are subject to extensive movement while in contact with other parts or surfaces and which are also subject to wear due to this movement shall be furnished with replaceable bearings, bushings or wear surfaces. These bearings (excluding the sealed type), bushings and wear surfaces shall have provisions for being lubricated with grease. All lubrication fittings shall comply with SAE J534. All lubrication fittings shall be easily accessible without having to remove components. Piping or flex hose may be used to extend lubrication fittings for access.

3.7.1 Lubrication plate. A lubrication plate or decal showing the following information shall be securely attached to each sweeper:

a. Point of application.

b. Service interval.

c. Type of lubricant.

d. Viscosity.

3.8 Special tools. Any special tools required for adjustments to support the sweeper's operation shall be furnished by the contractor.

3.9 Sound level. The sweeper exterior sound level shall not exceed 88 dB(A) when measured in accordance with the constant speed moving test of SAE J88. In lieu of operating the prime mover in an intermediate gear ratio as specified by SAE J88, the prime mover shall be operated at a forward speed of 35 miles per hour (mph) and the brush shall be operated at not less than 500 revolutions per minute (rpm). If the exterior sound level exceeds 84 dB(A), a noise hazard caution sign(s) shall be posted in accordance with paragraph 4.3 of MIL-STD-1474.

3.10 Performance. The snow broom shall be capable of sweeping a 11-feet wide swath at a broom angle of 45-degrees to the left and right of the centerline of travel at a speed of 35 mph minimum, in snow that is minimum of 2-inches deep. Additionally, the broom shall sweep 4-inches, minimum, deep snow

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at a minimum speed of 20 mph in the same manner. Density of the snow shall be 15 to 25 pounds per cubic foot. During each run, blower action shall leave the pavement bare to the surface. This performance pertains only to fair weather conditions with neglible winds.

3.10.1 Temperature conditions. The snow sweeper unit shall crank and operate at the rated capacity in ambient temperatures as low as -20°F with only the aid of engine glow plugs or measured shot ether injection system. The unit shall start, and operate at the rated capacity within 15 minutes, in ambient temperatures from -21°F to -40°F. Any equipment recommended by the engine and hydraulic system component manufacturers in order to meet this requirement, shall be furnished and installed on the unit. The entire unit shall suffer no damage by being stored in any ambient temperature to -65°F.

3.11. Components: Components of the sweeper shall consist of, but shall be not limited to, the following:

a. Brush assembly.

(1) Brush core.

(2) Brush frame.

(3) Brush hood.

(4) Hood air deflector and director.

(5) Brush angularity system.

(6) Brush drive system.

(7) Castoring system.

(8) Quick hitch.

b. Air blower.

(1) Blower mounting.

(2) Blower drive.

c. Engine skid assembly.

d. Hydraulic system.

e. Lighting system.

f. Brush and blower remote controls.

3.11.1 Brush assembly. The brush assembly shall consist of a cylindrically shaped core, mounting frame, hood and deflector, angling system, drive system, castors and hitch. The brush shall be filled with flat ring, continuous wafer type fillers. The fillers shall be in accordance with Code Numbers 01 and 03 of Military Specification MIL-F-83828 alternating (i.e. steel filler, plastic filler, steel filler, plastic filler, etc.) along the full length of the core. Spacers shall be either steel or plastic Code Number 04 or 05 of MIL-F-83828. The brush assembly shall be capable of being adjusted to provide different sweeping patterns by using a mechanical adjuster and shall lock into position

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when released. No tools shall be required to make any sweeping pattern adjustments.

3.11.1.1 Brush core. The brush core shall be 16 ± 1/4 feet minimum length, 46 +0 -1/16 inches diameter and shall be designed to facilitate bristle replacement, and to give maximum strength and balance. The core shall be bearing supported in the middle of the brush, and shall be designed to permit replacing filler lug channels in the event they need replacement.

3.11.1.2 Brush frame. The brush frame shall be welded steel construction designed to withstand the forces induced by airport snow removal. Frame components may be removable to allow for ease of brush replacement.

3.11.1.3 Brush hood. The brush hood shall cover at least 40 percent of the circumference of the brush. It shall be 14-gauge, minimum, steel hinged and welded or bolted to the brush frame. The hood shall include provisions for mounting a snow deflector on the front of the hood. It shall be a nonclog design to prevent the buildup of snow and ice under the hood at rated speeds. Brush hood shall provide the necessary quick access to brush for replacement of worn brush sections and inspections. Stops shall be provided to support the weight of the brush hood when it is opened back to facilitate brush replacement. Hydraulic cylinders, lines and fittings shall not be subjected to damage during brush assembly replacement.

3.11.1.4 Hood air deflector and director. A metal flap shall be mounted on the front edge of the hood to positively change the angle that the swept snow leaves the brush. Angular adjustment of the flap shall be by remote control hydraulically from the operator's cab.

3.11.1.5 Brush angularity system. The sweeper brush shall have a remote operator selectable angling capability. Brush shall swing 45-degrees right or left minimum travel from the transverse position, and shall have the necessary degrees of freedom to follow the pavement while sweeping either right or left at the rated sweeping speed. Brush shall sweep the same path whether angled right or left. Brush angling mechanism shall be hydraulically actuated and remotely controlled from the driver's seat. The brush head shall be self-leveling through use of parallel linkage with ball joints on each end between the angularity frames and the track mounting. The brush shall have stops to prevent it from coming in contact with the brush push frame when changing the angle from extreme left to extreme right or vice versa.

3.11.1.6 Brush drive system. The brush core shall be driven by a closed loop hydrostatic system either from both ends by a motor mounted on each outside end or by a single center mounted motor. Optimum brush power shall be available at an infinite number of brush speeds throughout a minimum range of 0 to 500 rpm. The drive motor(s) shall be protected within the brush framework. The pump shall be a heavy duty roller bearing variable displacement piston type.

3.11.1.7 Castoring system. The brush assembly shall be mounted on 16-ply minimum pneumatic tired, 360 degree revolving, single type castors having hydraulic or friction shimmy dampeners. Four of these castor assemblies shall be evenly spaced along the frame and mounted so that they shall support at least 70 percent of the weight of the broom assembly, and shall track within the sweeping path at all times. Castors which are in front of the brush or track outside the sweeping path shall be unacceptable. When adjusted according to the manufacturer's instructions, the castors shall operate on bare pavement without damage to the tire, and shall not bind, come into contact with the broom or rub against any other surface on the sweeper throughout their full rotational arc.

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All castor tires shall be filled with ARNCO "RE-PNEU" or an equivalent type polyurethane foam product.

3.11.1.8 Quick hitch. The brush assembly shall be designed for quick attachment and removal from the truck by means of a quick hitch device. The brush assembly shall be capable of being removed and installed, installation to include setting the sweeping pattern, by not more than two people within a maximum of 20 minutes for each operation. That portion of the quick hitch which is mounted on the truck shall in no manner interfere with normal truck operation when the brush assembly is detached. One complete brush portion of the quick hitch and either one or two complete truck portions of the quick hitch as specified (see 6.2), shall be furnished with each sweeper supplied on contract. The quick hitch shall be capable of sustaining all loads including side loads. It shall not induce bounce or oscillatory motion into the sweeper, and shall permit normal operational turning without binding or sweeper damage. There shall be retractable legs, or other means, for supporting the brush assembly when it is removed from the truck. It shall be supported at approximately the same inclination as when installed on the truck.

3.11.2 Air blower. The air blower shall be a double outlet centrifugal type driven by a variable displacement closed loop hydrostatic pump through a hydrostatic motor mounted directly to the fan impeller. Capacity of the fan shall be 10,000 cubic feet per minute (cfm) minimum. The maximum air velocity at each blower outlet shall be at least 325 mph, and both nozzles shall always be operating. The blower fan shall be variable speed throughout a minimum range of 25 to 2600 rpm. With the broom angled either left or right, one nozzle shall be in line with the leading edge of the brush head and the other nozzle shall be in line with the trailing edge to help dissipate the windrow. The nozzles shall not be located outside of the leading and trailing edges of the brush head. They may be located up to, but not more than, six inches inside of the leading and trailing edges of the brush head. The airblast shall be automatically changed when the brush is angled directing the airblast perpendicular to the direction of travel of the vehicle and toward the direction of brush discharge. The air discharge nozzles shall have a manual adjustment to direct the airblast to the outside of the sweeper unit for clearing runway lights. Additionally, the airblast shall have the ability of movement in a vertical area to allow cleaning the runway lights from top to bottom.

3.11.2.1 Blower mounting. The blower mounting shall be an adjustable quick disconnect type. The mounting shall be attached at the rear of the engine skid. When detached the blower and blower mounting shall not interfere with the normal operating functions of the truck. The blower mounting frame shall be equipped with two forklift tine channels of sufficient length, one on each side, so the assembly can safely be removed from or installed on the dump body. The channels shall completely surround the forklift tines to prevent dropping the assembly during removal or installation.

3.11.3 Engine skid assembly. The engine skid assembly shall consist of a commercial (see 6.5.1), water cooled, diesel engine having replaceable cylinder liners. The engine shall be mounted on a metal skid which will fit inside the bed of a standard 4 cubic yard (water level) dump truck body. The engine shall have a continuous horsepower rating of 190 minimum when rated in accordance with SAE J1349. The skid shall be designed to allow easy placement into or removal from the dump body by overhead crane or forklift. The skid assembly shall be equipped with two forklift tine channels, one on each side, as described in 3.11.2.1. The skid assembly, together with the blower assembly, shall be capable of being installed and removed by not more than 2 people within a maximum time of 45 minutes for each operation. A double hydrostatic pump drive

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shall be mounted to the engine's flywheel housing for direct drive of the brush and blower pumps. The engine shall have automatic high temperature and low oil cutoff devices and an electrical starting system. Instrumentation shall include, but shall not be limited to, an engine hourmeter, an ammeter, oil pressure gauge, fuel gauge, and coolant temperature indicator. The engine's air cleaner shall include a turbo type precleaner. The skid assembly shall include a fuel tank of 8 hours operating capacity. Means of securing the skid assembly within the dump body shall be provided by the supplier. The engine shall be supressed to limit electromagnetic radiation in accordance with SAE J551. The engine shall operate on JP-8 fuel conforming to MIL-T-83133 without detrimental effect to its components.

3.11.3.1 Cowling. The engine skid assembly, including the hydraulic components, shall be enclosed in a sheet metal cowling of 16 gauge minimum thickness. The cowling shall have removable access doors to facilitate servicing the engine and hydraulic components. The doors shall be securely held to the cowling by quick disconnect pins. The pins shall be attached to the cowling by chains or cables to prevent them from becoming separated from the sweeper. The entire cowling assembly shall be removable using common hand tools and shall be louvered as necessary to prevent overheating of the engine and hydraulics.

3.11.4 Hydraulic system. The hydraulic system shall consist of a reservoir, oil cooler, pump(s) and all necessary valves, including pressure release valves, and plumbing. The system shall supply fluid under pressure to the broom and blower drive circuits and to all hydraulically actuated components of the sweeper. The oil cooler shall be mounted in front of the diesel engine's radiator to cool the hydraulic oil to correct temperature under all conditions of use. The system shall include an automatic temperature control bypass to bring oil up to operating temperature. The hydrostatic reservoir shall be 15 gallons minimum with outside changeable filtering system. The tank shall have baffles to assist cooling and to eliminate aerating of the oil. An oil sight gauge and an oil temperature gauge shall be provided. An 8 micron absolute, full flow, bypass type filter shall be installed on the high pressure side of the loop. A 10 micron absolute, full flow, no bypass filter shall be installed on the low pressure side of the loop. Both filters shall incorporate restriction gauges or devices to indicate when replacement is necessary.

3.11.4.1 Hoses, tubing and couplings. All high pressure hoses shall be four or six spiral wire and shall be rated for a working pressure of not less than 5000 pounds per square inch (psi). Hoses from the pump to the brush frame shall be formed into an umbilical cord of the correct length for easy field installation. This umbilical cord shall be routed or supported in such a way as to prevent it from rubbing excessively against any part of the truck or sweeper. Hoses shall be equipped with male and female quick couplers for rapid removal/installation of the skid and brush assemblies. The quick couplers shall be of the fool proof design to prevent connecting the lines to the wrong coupling.

3.11.5 Lighting. Two auxiliary driving lights shall be mounted on the snow broom push frame near the quick hitch. The lights shall be connected to a weatherproof electrical quick disconnect terminal to facilitate installation/removal of the broom for sweeping operations. The lights shall be controlled from the remote control box. If the blower and/or skid assembly obscures the truck's rear lights, two auxiliary tail/stop/signal lights and marker lights shall be mounted on the rear of the blower or skid assembly such that they are visible when approaching the rear of the truck. The lights shall be furnished with hardware to allow connection to the dump truck's lighting

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system. Not less than 2 working lights shall be furnished and mounted to illuminate the entire engine area at night so personnel can check/service fluid levels and inspect the engine assembly. The switch to control the lights shall be mounted on the remote control box. A warning beacon shall be mounted on the engine skid assembly in a position such that it is visible from 3600. The beacon shall be a class 2 flashing type lamp in accordance with SAE J845, with a yellow globe conforming to SAE J578. The beacon shall emit at least 90 flashes per minute. The control switch for the beacon shall be mounted on the remote control box.

3.11.5.1 Battery. The battery shall be the sealed maintenance free type. A battery box with cover shall be provided and mounted in a location to provide easy access for checking/replacement.

3.11.6 Brush and blower remote controls. An umbilical cord and control box shall allow the operator complete control of the sweeping and blowing unit from his position inside the cab. A quick disconnect coupler shall be furnished to be permanently mounted through the right rear corner of the vehicle cab. The coupler shall be the multipin type, having sufficient pins to allow the instrumentation specified herein. The coupler shall be designed to connect to the umbilical cord from the engine skid assembly on the outer side of the cab and to the umbilical cord from the control box on the inner side. The coupler shall have a hinged weatherproof cover on the outer side, and the installation of the coupler through the cab wall shall prevent the entrance of dust and moisture into the cab. The umbilical cord from the engine skid to the coupler and from the coupler to the control box shall be the correct length for the vehicle(s) specified (see 3.12). As a minimum, the control box shall have the following instrumentation:

a. Hydrostatic motor fluid pressure gauge.

b. Hydraulic fluid temperature gauge

c. Auxiliary engine, brush and blower speed gauges.

d. Engine start switch, with key.

e. Engine oil pressure warning light.

f. Engine coolant temperature warning light.

g. Speed controls for engine, brush and blower.

h. Control for raising/lowering the brush.

i. Control for angling brush.

j. Control for brush hood air deflector position.

k. Manual switch for vertical oscillation in cleaning snow from runway lights and to center blower ducts for transport

l. Switches to operate the warning beacon, auxiliary rear lights, auxiliary driving lights and engine skid assembly working lights.

The control box shall be lighted for night operation. All gauges, controls and switches shall be logically arranged and grouped so that all functions for the broom are grouped together, all functions for the blower are grouped together,

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auxiliary light switches are grouped together, etc. The remote control box shall be completely enclosed and weatherproof. The remote control box shall be fitted with a removable weatherproof sealed type cover to protect the controls when exposed to the elements.

3.12 Installation. All parts necessary for attaching the sweeper assembly to the makes and models of dump trucks contained on the list furnished by the Government (see 6.2.1) shall be furnished. The installation will be performed by Air Force personnel at base level maintenance.

3.12.1 Installation instructions. Installation instructions for mounting the entire sweeper assembly, using the contractor supplied components, shall be included in the Air Force Technical Order (TO). There shall be separate instructions for each make and model of dump truck covered by the contract.

3.14 Workmanship. Workmanship of the sweeper unit, accessories and attachments shall be of the quality prevailing among manufacturers normally producing equipment of the type specified herein. Defective components or parts and assemblies which have been repaired to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction shall be in accordance with the highest standards of industry. All welders employed in the fabrication of the sweeper, except for commercial (see 6.5.1) components, shall be certified in accordance with the applicable requirements of the American Welding Society.

4. QUALITY ASSURANCE.

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use any inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that the sweeper conforms to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the sweeper shall be classified as follows:

a. First article testing (see 4.4 and 6.3).

b. Acceptance testing (see 4.5).

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

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4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Engine speed.

f. Engine oil pressure.

g. Engine coolant temperature.

h. Engine hourmeter reading at the start and finish of each test.

i. Hydraulic fluid pressure at the pump outlet.

j. Hydraulic fluid temperature in the reservoir.

4.3.3 Preparation for tests. The sweeper, equipped with all specified equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. Throughout all tests specified herein, the sweeper shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any operating condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between sweeper components or between the sweeper and the ground, with the exception of normal contact by the tires and broom.

f. Misalignment of components.

g. Evidence of undesirable mobility characteristics, including instability in handling during cornering, braking, etc.

h. Conditions which present a safety hazard to personnel during operation, service, or maintenance.

i. Overheating of the engine or any other sweeper component.

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4.4 First article testing.

4.4.1 First article test samples. The first article test samples shall be identical to the items procured under the terms of the contract. Approval of the first article test report shall not constitute a waiver by the Government of its rights under the provisions of the contract. At the manufacturer's option, either one or two first article samples shall be produced and tested as follows:

a. Contractor test sample (see 4.4.2.1).

b. User test sample (see 4.4.2.2).

4.4.2 First article tests.

4.4.2.1 Contractor test sample. The contractor test sample shall be subjected to the tests described in 4.6.1 through 4.6.11.

4.4.2.2 User test sample. The user test sample shall successfully complete the tests in 4.6.1 and 4.6.2 prior to being shipped to the user test facility for the test described in 4.6.12. When the manufacturer chooses to produce only one first article test sample, it shall successfully complete all tests described in 4.6.1 through 4.6.11 prior to being shipped to the user test facility for the test described in 4.6.12.

4.4.3 Test report. Upon completion of the first article tests, a first article test report shall be prepared in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets and the report. The following shall be incorporated as a part of the test report:

a. All data collected, failures, maintenance performed, and other events recorded shall be identified by accumulated operation time, miles, cycles, or position in the test procedure, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made, or proposed to be made and the predicted effectiveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. List of special tools and diagnostic equipment necessary to perform field level maintenance.

4.5 Acceptance testing. Each production sweeper shall be subjected to the examination specified in 4.6.1 and the tests specified in 4.6.2 and 4.6.11.

4.6 Test methods.

4.6.1 Examination. Each complete snow sweeper and attachments shall be

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examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with special requirements shall receive particular attention. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements shall be cause for rejection.

4.6.2 Run-in test. All snow sweeper units produced in fulfillment of the contract or order shall be completely assembled, adjusted, lubricated and otherwise serviced for operation. A run-in test shall be conducted in accordance with the manufacturer's recommendations. During this test all controls, adjusting mechanisms, hydraulic systems and other accessories shall be operated sufficiently to ensure against leaks, restrictions or malfunctions. Failure to pass any phase of this test shall be cause for rejection.

4.6.3 Sound level test. The sweeper unit shall be mounted on a dump truck chassis furnished by the manufacturer and tested as specified in 3.9 to determine conformance to the requirements therein.

4.6.4 Sweeping endurance test. The sweeper shall be operated for a minimum of 100 hours in the sweeping mode. At least 75 percent of the total time shall be for periods of not less than 4 hours of continuous operation at a forward speed of not less than 30 mph and at a brush speed of not less than 450 rpm. The sweeping pattern shall be set and maintained at the manufacturer's maximum recommended value throughout the test. During this test the sweeper engine shall be operated solely on JP-8 turbine fuel in accordance with MIL-T-83133. Jet A-1 turbine fuel in accordance with ASTM D1655 may be used in lieu of JP-8.

4.6.5 Temperature test. The manufacturer shall submit notarized certification that the sweeper unit offered by him will meet or exceed the cranking, operating, and storage requirements specified in 3.10.1.

4.6.6 Quick hitch test. The brush assembly shall be attached and detached from the truck to demonstrate the ease of coupling and uncoupling and to verify the time requirement of 3.11.1.8. During uncoupling, and prior to recoupling, of the brush assembly, the truck shall be backed away from the brush assembly such that it is not aligned with brush hitch. The recoupling portion of the test shall start with the truck in this position. While the broom is coupled to the truck and in the sweeping mode, it shall be operated for a distance of 2000 feet at speeds of 20 to 35 miles per hour. Note shall be taken of any pronounced bouncing or oscillatory motion of the sweeper head. Turns to the extreme right and left shall be made with the vehicle. Any binding or sweeper damage shall be cause for rejection.

4.6.7 Air blower test. The air blower shall be operated to demonstrate that it meets the requirements specified in 3.11.2. Backup in the form of engineering data shall ensure the minimum capacity of 10,000 cfm of the blower fan.

4.6.8 Engine certification. The manufacturer shall submit notarized certification, backed by performance curves and engineering data, that the engine and fuel tank meets or exceeds the requirements of 3.11.3.

4.6.9 Assembly test. The engine skid and blower assemblies shall be removed and installed to verify the time requirement of 3.11.3. The engine skid and blower shall be removed from the truck and placed on the ground using a forklift. They shall then be installed on the truck using the forklift. Installation shall include making all hydraulic and electrical connections and

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adjustments necessary for operation.

4.6.10 Maintainability demonstration. Compliance with the repair time requirements shall be demonstrated by task performance. All tasks listed in 3.4.2 shall be performed.

4.6.11 Inspection of preservation, packaging, packing and marking for shipment and storage. Preservation, packaging, packing and marking for shipment and storage shall be inspected to verify conformance with the terms of the contract or order.

4.6.12 User test. The sweeper shall be delivered to the location specified (see 6.2) and shall be available for 45 days of operational use, or 60 calendar days; whichever comes first. A "day of operational use" shall be defined as any eight hour period specified by the Government.

4.6.12.1 Availability and operation. The test period shall begin after the sweeper has been inspected and prepared for use. The sweeper shall be available for any eight hour period specified by the Government each day, seven days a week for a total of 360 operating hours. The sweeper may be used for up to three consecutive eight hour periods during a 24 hour calendar day. The Air Force shall notify the manufacturer a minimum of 12 hours in advance if it anticipates using the sweeper for two or more consecutive eight hour periods in a 24 hour calendar day. Any period of more than four hours of the same eight hour "operational day" that the sweeper is out of service for maintenance or repair shall not be considered an available day of the test period. Test operations shall simulate normal Air Force usage and shall not exceed any operating parameters defined by this specification. The sole criteria for acceptance/rejection of the user test shall be based on the performance requirements of this specification.

4.6.12.2 Maintenance during user test. The sweeper shall be operated and scheduled maintenance shall be performed by Air Force personnel using the operation and maintenance procedures contained in the preliminary technical data to be supplied with the sweeper. The Air Force shall provide diesel fuel and consumables such as air and water and also shop space for any corrective maintenance required during the test. The Air Force shall notify the manufacturer any time corrective maintenance is required. Any parts or corrective actions required shall be provided by the manufacturer. The manufacturer shall notify base vehicle maintenance personnel before performing corrective maintenance and shall allow them to observe while the maintenance is being performed.

4.6.12.3 User test log. The manufacturer shall maintain a daily log, with an entry made for each calendar day of the user test. Each calendar day entry shall be broken down into three eight-hour periods, with any pertinent data or actions occurring during the period being recorded. Any corrective maintenance performed shall be recorded, including the date and time that the sweeper was removed from service and the date and time that it was returned to service. The Air Force shall designate one or more responsible personnel to sign the log at the end of each eight-hour period. The log may be signed during the next duty day if a responsible person is not available at the completion of a particular eight-hour period. The log shall be made part of the first article test report. The manufacturer is responsible for recording all data required for the first article test report.

5. PACKAGING.

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5.1 Preparation for delivery. Preparation for delivery shall be in accordance with the terms of the contract.

6. NOTES.

6.1 Intended use. The sweeper unit specified herein is designed to be mounted on a dump truck and used to sweep runways clear of snow or slush at high speed.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) When a first article sample is required (see 3.1 and 6.3).
(d) Whether one or two truck portions of the quick hitch are required (see 3.11.1.8).
(e) Location of the user test (see 4.6.12).

6.2.1 List of applicable trucks. At the time of invitation for bid or request for proposal, the Government shall provide a list of all applicable makes and models of dump trucks currently in its inventory. The intent of this list is to provide prospective bidders with the range of vehicles on which the sweeper could be installed so that they may determine the necessary hardware and formulate prices for the construction of quick hitch kits. After a contract has been awarded to the successful bidder, the Government shall identify the specific trucks on which the sweepers shall be installed. The trucks shall be identified by manufacturer, model year and model number.

6.3 First Article Unit. The sample may be production item or a preproduction item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed item and shall be made or shall have been made by the methods that are to be used or were used in production of items to fulfill the terms of the contract. Approval of a First Article Unit shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification. The First Article Unit may be shipped as part of the contract.

6.3.1 Consideration of data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Requirements List (CDRL), incorporated into the contract. When the provisions of DAR FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
4.4.1, 4.4.3, 4.6.1, 4.6.2, 4.6.3, 4.6.4, 4.6.5, 4.6.6, 4.6.7, 4.6.8, 4.6.9, 4.6.10, 4.6.12, 4.6.12,1, 4.6.12.2, 4.6.12.3.	First Article Inspection Report	DI-NDTI-80809A

(Data item descriptions related to this specification, and identified in section

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6 will be approved and listed in DOD 5000.19L, Vol II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 Subject Term (Key Word) Listing.

Blower
High Speed
Push type
Rotary
Snow
Sweeper

6.5 Definitions.

6.5.1 Commercial. A manufacturer's standard (product or process) having been marketed, in production, and sold in substantial quantities to the general public for a minimum of one year prior to the date of invitation for bid or request for proposal. An item is "sold in substantial quantities" only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities, such as models, samples, prototypes, or experimental units, do not meet this requirement. The "general public" is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification, items acquired for "Government end use" include items acquired for foreign military sales.

MILITARY INTERESTS:

Custodians

Air Force - 99

PREPARING ACTIVITY:

Air Force - 84

Review Activities

Navy - YD

Project No. 3825-0210

FSC 3825

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-83829	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
Sweeper, Rotary, Snow, Front Mounted

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME WR-ALC/LVRG	b. TELEPHONE (Include Area Code) (1) Commercial: (912) 926-2161 (2) AUTOVON: 468-2161
c. ADDRESS (Include Zip Code) 225 Ocmulgee Court Robins AFB GA 31098-1647	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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